

Test of the accuracy of smallcurvature and minimumenergy reference paths for parametrizing the search for leastaction tunneling paths: (H,D)+H'Br→(H,D)Br+H'

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Test of the accuracy of small-curvature and minimum-energy reference paths for parametrizing the search for least-action tunneling paths: $(\text{H,D}) + \text{H}'\text{Br} \rightarrow (\text{H,D})\text{Br} + \text{H}'$

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The least-action semiclassical algorithm for multidimensional tunneling probabilities [B. C. Garrett and D. G. Truhlar, *J. Chem. Phys.* **79**, 4931 (1983)] has usually been employed by interpolating tunneling paths between two limits, a least-motion limit appropriate for large reaction-path curvature and a minimum-energy limit appropriate for small reaction-path curvature. In the present study we test whether, when the reaction-path curvature is small, more accurate results might be obtained by using a general small-curvature reference path. Least-action algorithms with both types of reference paths are compared to each other, to five other semiclassical approximations, and to accurate quantal dynamical rate constants for one three-dimensional and two collinear reactions with the mass combination $\mathcal{L} + \mathcal{H} \mathcal{L}' \rightarrow \mathcal{L} \mathcal{H} + \mathcal{L}'$ where $\mathcal{L}$ and $\mathcal{L}'$ denote light atoms (H or D) and $\mathcal{H}$ denotes a heavy atom (Br). We find, perhaps surprisingly, that the usual least-action method works best. This is encouraging because the minimum-energy reference is easier than the small-curvature reference to extend to polyatomic reactions.

I. INTRODUCTION

Variational transition state theory (VTST) has proven to be a very useful semiclassical approximation for bimolecular gas-phase reaction rates.¹⁻³ Several methods have been used to incorporate tunneling into VTST.¹⁻³ However, tunneling approximations are very difficult to test against experimental results because when discrepancies arise, it is very hard to know if they are due to the dynamical approximations or the potential-energy surface or both. Accurate quantum dynamical calculations, when available, alleviate this problem because they are performed using analytic potential-energy surfaces which can also be used in VTST calculations. Comparison of accurate and approximate dynamical calculations based on the same potential function removes the uncertainties in evaluating the approximate theory. The difficulty of carrying out accurate quantum-mechanical calculations limits the availability of such results. Thus many checks have been carried out by treating the quantum dynamics in a reduced-dimensionality (collinear) world, and carrying out the approximate calculations under the same restriction for comparison. Recently, accurate quantal rate constants were used to test the validity of VTST and semiclassical tunneling calculations for eight isotopomeric analogs of the $\text{H} + \text{H}_2$ and $\text{O} + \text{H}_2$ reactions.⁴ Another system for which accurate three-dimensional rate constants are available⁵ is



for a DIM-3C (Refs. 6 and 7) potential-energy surface. In the present article we use these accurate results to test VTST and several different semiclassical tunneling algorithms. A more limited test was performed previously⁸ for the collinear analog of reaction (R1) and for reactions (R1) and



in three dimensions. However, the three-dimensional quantal results available at the time contained approximations which have since⁵ been removed. Furthermore, we have developed more accurate semiclassical tunneling methods.⁹⁻¹¹ These developments, plus our interest in another reaction, $\text{H} + \text{DCI} \rightarrow \text{HCl} + \text{D}$, which is treated in the following paper,¹² have prompted us to reexamine the accuracy of various approximate methods for reactions (R1) and (R2). Our goal is to determine which of the presently available semiclassical tunneling methods yields the greatest accuracy when used with VTST.

This work was initiated because of questions which arose in attempting to predict the exchange rate constant for the $\text{H} + \text{DCI}$ reaction mentioned above. Recently our research group developed two new global potential-energy surfaces for the H_2Cl system, and VTST calculations were performed on these surfaces.¹² However, it was not clear which method would be most accurate for the tunneling calculations. In general, the accuracy of the various tunneling methods depends on the curvature of the reaction path, which in turn depends strongly on the mass combination. The mass combination of a very heavy atom being transferred between two hydrogens is clearly one extreme, and in fact it leads to unusually small reaction-path curvature. However, there are no accurate quantum-mechanical rate coefficients for $\text{H} + \text{DCI}$, and previous comparisons^{5,8,10,11} for the Br analog, which also represents the extreme limit, did not include both possible reference paths⁸ in the tunneling calculations. As a result, one is uncertain which method is most reliable to predict experimental rates. The present study of a very similar reaction with a very similar mass

us which tunneling method is most reliable for a case like this.

In this paper we test approximate rate coefficients against accurate quantum dynamics for reactions (R1) and (R2) in a collinear world and reaction (R1) in three dimensions. The approximate calculations are based on improved canonical variational transition state theory¹³ (ICVT) with several^{9-11,13-15} different tunneling approximations. All calculations use the optimized DIM-3C surface.^{6,7,16}

Section II summarizes variational transition state theory and the semiclassical tunneling methods. Sections III and IV present results and discuss and compare the various tunneling methods. Section V summarizes the main conclusions.

II. METHODS

The calculations were performed using conventional transition state theory^{17,18} (TST) and improved canonical variational transition state theory^{13,18} (ICVT) with small-curvature, large-curvature, and least-action ground-state tunneling approximations. The versions of the ground-state transmission coefficients used were the minimum-energy-path semiclassical adiabatic ground-state approximation¹³ (MEPSAG), the small-curvature semiclassical adiabatic ground-state approximation^{14,15} (SCSAG), the least-action ground-state approximation¹⁰ (LAG), the least-action ground-state approximation with a small-curvature reference path¹⁰ [LAG(SC)], and the large-curvature ground-state approximation, in both the original version⁹ (LCG) and version 3^{10,18,19} (LCG3).

A. Variational transition state theory

The calculations were performed as described elsewhere.^{11,13-15,19-21} Bending vibrations were treated by a harmonic-quartic approximation with force constants fit to reproduce the actual bend potential at 10 and 20 deg. The stretching vibrations are treated by a WKB approximation.¹¹ The rotational partition functions were determined by the quantum-mechanical rigid rotor approximation, as usual.¹³

In the low-temperature limit (0 K) the variational transition state is located at the maximum of the vibrationally adiabatic ground-state potential curve $V_a^G(s)$ given by

$$V_a^G(s) = V_{\text{MEP}}(s) + \epsilon^G(s). \quad (1)$$

In Eq. (1), s is the distance along the minimum energy path (MEP) measured from the saddle point, which defines $s = 0$, $V_{\text{MEP}}(s)$ is the potential along the MEP at a point s , and $\epsilon^G(s)$ is the sum of the zero-point energies of the modes orthogonal to the reaction path at a distance s from the saddle point (there is one such mode, a stretch, for the collinear case, and two more, a doubly degenerate vibration, in three dimensions). The classical barrier height $V^\ddagger$ is the value of $V_{\text{MEP}}(s)$ at the saddle point. The maximum value of the adiabatic ground-state potential at s is denoted $V_a^{\text{AG}}(s)$, and we also define

$$\Delta V_a^{\text{AG}}(s) = V_a^{\text{AG}}(s) - \epsilon^{\text{RG}}, \quad (2)$$

where ϵ^{RG} is the zero-point energy of the reactants.

Quasiclassical rate constants (sometimes called hybrid rate constants) are obtained by quantizing the bound degrees of freedom of the generalized transition states but treating the reaction coordinate motion as classical. The quantal effects on the reaction coordinate are included by a multiplicative transmission coefficient $\kappa^{\text{ICVT/YG}}(T)$ such that the final rate constant is

$$k^{\text{ICVT/YG}}(T) = \kappa^{\text{ICVT/YG}} k^{\text{ICVT}}(T), \quad (3)$$

where $k^{\text{ICVT}}(T)$ is the quasiclassical rate constant. The transmission coefficient accounts for dynamical quantum effects of reaction coordinate tunneling, nonclassical reflection, and coupling of reaction-coordinate tunneling motion with vibrational motions. The final rate constant is abbreviated as ICVT/YG where Y stands for any one of the semiclassical tunneling methods mentioned above and G stands for the ground-state method^{13,18} of calculating the transmission coefficient.

B. Semiclassical tunneling approximations

The skew angle β for a system is defined as the angle between the entrance and exit axes in the coordinate system^{17,20} that diagonalizes the local kinetic energy with a constant reduced mass. For the reactions under consideration here it is given by

$$\beta = \arccos(m_{\text{H}}/m_{\text{HBr}}) \quad (4)$$

for (R1) and

$$\beta = \arccos(m_{\text{H}}m_{\text{D}}/m_{\text{HBr}}m_{\text{DBr}})^{1/2} \quad (5)$$

for reaction (R2). The skew angle is a good measure of the reaction-path curvature, and in fact there is a rigorous integral relation between $\cos \beta$ and reaction-path curvature.¹⁸ The large skew angles obtained from Eqs. (4) and (5) indicate relatively small reaction-path curvature for reactions (R1) and (R2) and they indicate in a general way which semiclassical tunneling methods are well suited to these systems. (However, as shown in the present study, the skew angle alone is not sufficient to determine the best tunneling method.)

Six different approximations are applied for calculating the transmission coefficient in this paper. Two semiclassical adiabatic ground-state (SAG) methods are based on the calculation of one-dimensional action integrals along the reaction path. MEPSAG is the simplest method and is based on the assumption that all tunneling occurs along the MEP. This is valid only if the system experiences no positive or negative internal centrifugal forces causing the center of the vibrational probability distribution to deviate from the MEP. However, when a tunneling system has a curved reaction coordinate it experiences negative internal centrifugal effects which lead to an optimum tunneling path on the concave side of the MEP.^{15,22-24} The SCSAG method accounts for this effect by using a curvature-dependent reduced mass to evaluate the imaginary-action integral, while retaining the vibrationally adiabatic approximation for the effective potential along the path. This is valid only when reaction-path curvature is small so that the dominant tunneling path lies within one vibrational half-amplitude of the MEP, i.e., within the distance from the MEP to the vibrational turning point. When reaction-path curvature is larger, one must di-

rectly evaluate the nonadiabatic effective potential along the dominant tunneling path or a good approximation to it. The large-curvature ground-state models (LCG and LCG3) do this by approximating the dominant tunneling path as a straight line in mass-scaled Cartesian coordinates. The termini of the straight line are located on a reference path in the vibrationally adiabatic region at or on the classically allowed side of the reaction-coordinate turning points $s_>$ and $s_<$ on the $V_a^G(s)$ potential curve. In the original version (LCG), the reference path is chosen as a small-curvature-type path, whereas in version 3 (LCG3), the MEP is used as the reference path. [The MEP has two advantages over the SC reference path: (1) it is more convenient; and (2) it allows a wider range of possible tunneling paths.] An energy-dependent quasiclassical vibrationally adiabatic distribution function is used to weight tunneling paths initiating in the interval $(-\infty, s_<)$ and ending in the interval $(s_>, \infty)$.

The least-action ground-state (LAG) tunneling method was developed to treat chemical reactions which have optimum tunneling paths that lie between the small- and large-curvature limiting paths. The LAG method provides a way of treating a sequence of trial paths in a consistent manner and evaluating which path in this sequence is most optimum, i.e., has the least imaginary action, leading to the least exponential decay so it dominates the tunneling. The sequence of paths for an available energy is specified by a set of trial paths depending on a single parameter α (with $0 \leq \alpha \leq 1$) such that $\alpha = 0$ yields the reference path and $\alpha = 1$ yields the large-curvature tunneling path. The optimum tunneling path depends on the path termini, and again an energy-dependent vibrationally adiabatic distribution is used to weight paths with various termini. For a given pair of tunneling path termini, which lie on the reference path, the optimum tunneling path will be the path from the one-parameter sequence with the most tunneling, and the optimum value of α is called α_{opt} . The LAG method employs the minimum-energy path as the reference path, as in the LCG3 method. Then the tunneling path becomes the MEP in the limit of zero curvature and the same as in the LCG3 method in the limit of large curvature. The LAG(SC) method differs from the LAG method in the choice of the $\alpha = 0$ reference path; in particular, the LAG(SC) method uses the small-curvature-type reference path of the original LCG method for $\alpha = 0$. The LAG and LAG(SC) methods should both provide good approximations for calculating tunneling probabilities for chemical reactions with small, medium, and large curvature of the reaction path.

The original motivation for using the MEP as a reference path in the LAG and LCG3 methods was simplicity. In particular, the small-curvature-type reference path of the LCG and LAG(SC) methods are hard to generalize to arbitrary polyatomic systems. But one might ask how big of a penalty one pays for this simplicity or whether the greater range of paths allowed by using the MEP as the reference path might be an advantage. Although the LAG method has been very successful for a wide variety of systems, one might ask whether, when the reaction-path curvature really is small, the LAG(SC) method might be more accurate because of its explicit use of a small-curvature-type path or

TABLE I. Saddle point and adiabatic ground-state maximum properties for collinear HBrH and DBrH on the DIM-3C potential-energy surface.

Species	HBrH	DBrH
Conventional transition state		
$R_{AB}(a_0)$	2.99	2.99
$R_{BC}(a_0)$	2.99	2.99
$V^\ddagger$ (kcal/mol)	5.49	5.49
$\hbar\omega_{\text{str}}(\text{cm}^{-1})$	1794	1525
$\hbar\omega^\ddagger(\text{cm}^{-1})$	1281i	1073i
$\hbar\omega_{\text{bend}}(\text{cm}^{-1})$	358	312
Adiabatic ground-state maximum		
$R_{AB}(a_0)$	2.99	3.01
$R_{BC}(a_0)$	2.99	2.96
$V_a^{\text{AG}}(\text{kcal/mol})$	8.03 ^a	7.66 ^a
$\Delta V_a^{\text{AG}}(\text{kcal/mol})$	4.29 ^a	3.92 ^a
$\hbar\omega_{\text{str}}(\text{cm}^{-1})$	1794	1546
$\hbar\omega_{\text{bend}}(\text{cm}^{-1})$	358	312

^a Collinear reaction.

whether the LAG method is more accurate because of the greater range of possible paths. That is the question that we came up against in our recent studies of the $\text{H} + \text{DCI} \rightarrow \text{HCl} + \text{D}$ reaction,¹² and the calculations in the present paper were therefore undertaken to answer this question.

In discussing our results below, we will find it informative to show tunneling paths at a typical tunneling energy. For this purpose we use the most probable LAG tunneling energy, $E_{\text{mp}}^{\text{LAG}}(T)$, defined as the energy at which the integrand peaks in a thermal average over fixed-energy tunneling probabilities $P^{\text{LAG}}(E)$, i.e., the energy where $P^{\text{LAG}}(E)\exp(-E/k_B T)$ maximizes, where E is the energy, k_B is Boltzmann's constant, and T is temperature.

III. CALCULATIONS

A. Potential-energy surface and vibrationally adiabatic maxima

The potential-energy surface is based on the DIM-3C formalism as described by Baer and Last.^{6,7} The g parameter is given the value 0.22. Table I summarizes the properties of

TABLE II. Accurate rate constants for collinear reaction (R1) (k_1^{1D}), three-dimensional reaction (R1) (k_1), and collinear reaction (R2) (k_2^{1D}).

$T(\text{K})$	k_1^{1D} ($\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$)	k_1 ($\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$)	k_2^{1D} ($\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$)
150	1.2(1)	1.3(–16)	5.5(0)
200	3.8(1)	7.6(–16)	2.3(1)
250	1.1(2)	3.4(–15)	7.3(1)
300	2.5(2)	1.1(–14)	1.9(2)
350	5.0(2)		3.8(2)
400	8.9(2)		7.0(2)
450	1.4(3)		1.1(3)
500	2.1(3)		1.7(3)
600	4.0(3)		3.2(3)

^a These results, from Refs. 5 and 8, are the denominators for the ratios in Tables III–V and may be used, if desired, to convert the ratios in subsequent tables to rate constants.

^b $1.2(1) \equiv 1.2 \times 10^1$.

TABLE III. Ratio of approximate rate constants to accurate rate constants for collinear H + BrH.

$T(K)$	$\dagger$	ICVT	ICVT MEPSAG	ICVT SCSAG	ICVT LAG	ICVT LAG(SC)	ICVT LCG	ICVT LCG3
150	0.0021	0.0021	0.34	2.5	1.2	2.5	0.56	1.1
200	0.028	0.028	0.45	1.8	0.88	1.9	0.78	0.85
250	0.094	0.094	0.52	1.4	0.74	1.4	0.84	0.72
300	0.19	0.19	0.61	1.2	0.74	1.2	0.89	0.73
400	0.37	0.37	0.71	1.0	0.76	1.1	0.91	0.76
500	0.52	0.52	0.78	1.0	0.81	1.0	0.94	0.81
600	0.62	0.62	0.82	0.97	0.84	0.99	0.94	0.84

the surface at the saddle point and the maximum of the ground-state adiabatic curve for both surfaces for both reactions (R1) and (R2). The HBrH adiabatic ground-state barrier height in three dimensions relative to reactants is 1.06 kcal/mol higher than the adiabatic barrier for the collinear case, which raises V_a^{AG} to 9.09 kcal/mol and ΔV_a^{AG} to 5.35 kcal/mol.

B. Rate constants

Table II gives the accurate rate constants for collinear (R1) and (R2),⁸ and for three-dimensional reaction (R1).⁵ Ratios of the present semiclassical rate constants to these accurate rate constants for collinear reactions (R1) and (R2) and for three-dimensional reaction (R1) are presented in Tables III, IV, and V, respectively. Figures 1 and 2 are semilog plots of the transmission probabilities versus energy for collinear reaction (R1). Similar plots are given for three-dimensional reaction (R1) in Figs. 3 and 4, and for collinear reaction (R2) in Figs. 5 and 6.

C. Dominant tunneling energies and tunneling paths

Figure 7 is a contour plot of potential energy with superimposed minimum-energy and tunneling paths. The figure

employs a mass-scaled coordinate system which is defined for an A + BC system as

$$x = R_{AB} + \left(\frac{m_C}{m_C + m_B} \right) R_{BC},$$

$$y = \left(\frac{\mu_{BC}}{\mu} \right)^{1/2} R_{BC},$$

where μ_{BC} is the reduced mass of the BC pair, μ is the reduced mass of relative translation of A with respect to B, and R_{AB} and R_{BC} are interatomic distances. This is one special case of a coordinate system that diagonalizes the kinetic energy with the same reduced mass for each square term, and so—as discussed above—the angle between the entrance and exit valleys is β .²⁰

Figure 7 shows the reaction-coordinate classical turning points, determined by $E = V_a^G(s_<) = V_a^G(s_>)$, for the dominant energy of tunneling and the straight line tunneling path for collinear reaction (R1) at 300 K, and similarly for reaction (R1) in three dimensions at 200 K. Tables VI and VII contain, for reaction (R1) in one dimension at 300 and 400 K and three dimensions at 200 and 300 K, the dominant tunneling energy, the reaction-coordinate turning points, and the optimum α . The distances $s_<$ and $s_>$ are measured in the coordinate system just defined.

TABLE IV. Ratio of approximate rate constants to accurate rate constants for collinear D + BrH.

$T(K)$	$\dagger$	ICVT	ICVT MEPSAG	ICVT SCSAG	ICVT LAG	ICVT LAG(SC)	ICVT LCG	ICVT LCG3
150	0.012	0.011	0.46	2.2	0.96	2.2	0.55	0.95
200	0.086	0.083	0.57	1.6	0.79	1.6	0.76	0.79
250	0.22	0.21	0.68	1.3	0.79	1.4	0.88	0.79
300	0.34	0.33	0.73	1.2	0.79	1.2	0.90	0.79
400	0.55	0.54	0.83	1.1	0.86	1.1	0.96	0.86
500	0.68	0.67	0.88	1.0	0.90	1.1	0.98	0.90
600	0.77	0.76	0.92	1.0	0.93	1.0	0.99	0.93

TABLE V. Ratio of approximate rate constants to accurate rate constants for three-dimensional H + BrH'.

$T(K)$	$\dagger$	ICVT	ICVT MEPSAG	ICVT SCSAG	ICVT LAG	ICVT LAG(SC)	ICVT LCG	ICVT LCG3
150	0.0030	0.0030	0.40	3.3	2.6	3.4	0.55	1.4
200	0.042	0.042	0.59	2.4	1.5	2.5	0.93	1.1
250	0.14	0.14	0.70	1.8	1.1	1.9	1.1	0.98
300	0.26	0.26	0.79	1.6	0.99	1.6	1.1	0.95

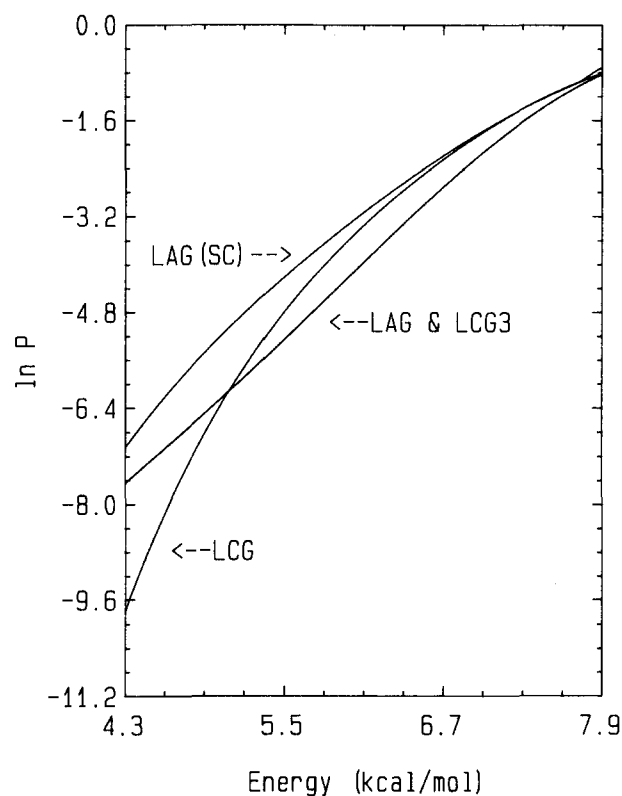


FIG. 1. Transmission probabilities as a function of total energy for the collinear reaction $\text{H}' + \text{BrH} \rightarrow \text{H}'\text{Br} + \text{H}$. Throughout this paper, total energy E is defined relative to reactants at infinity with the diatomic at its equilibrium distance.

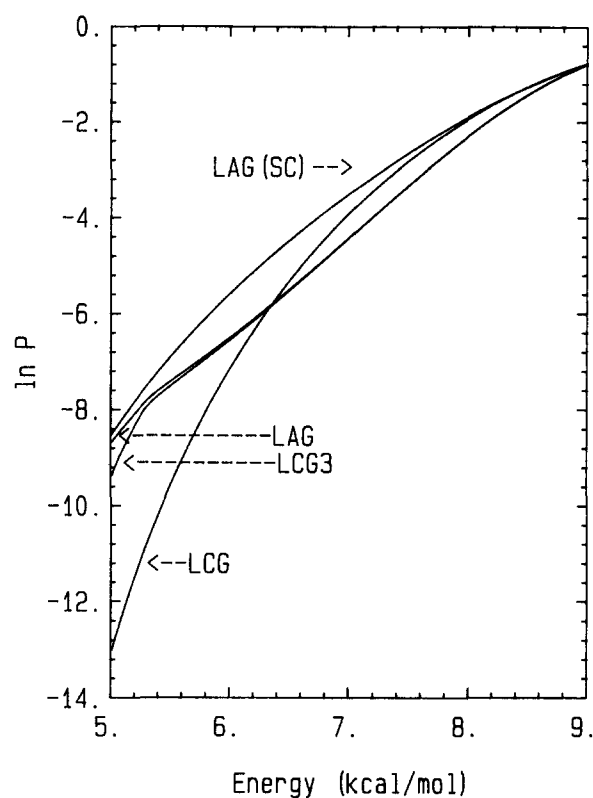


FIG. 3. Same as Fig. 1 for the three-dimensional case.

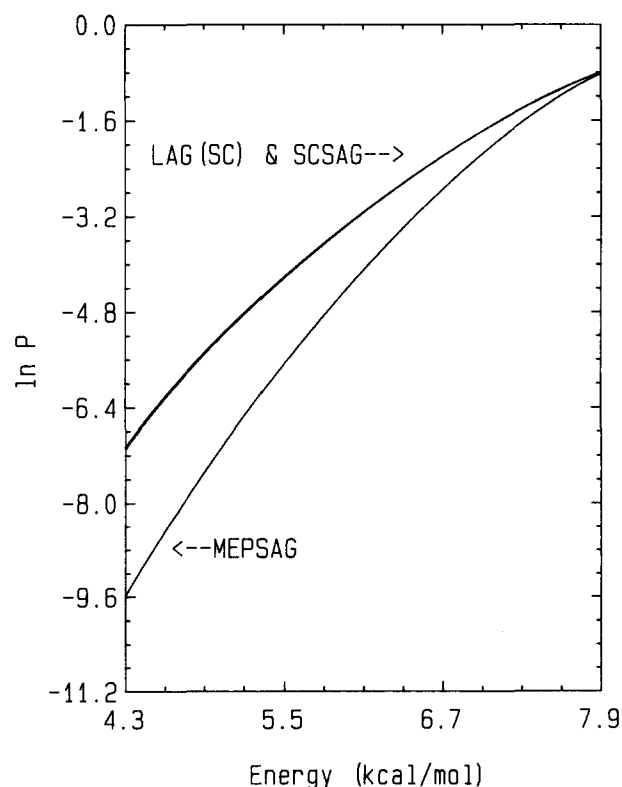


FIG. 2. Same as Fig. 1 for other methods.

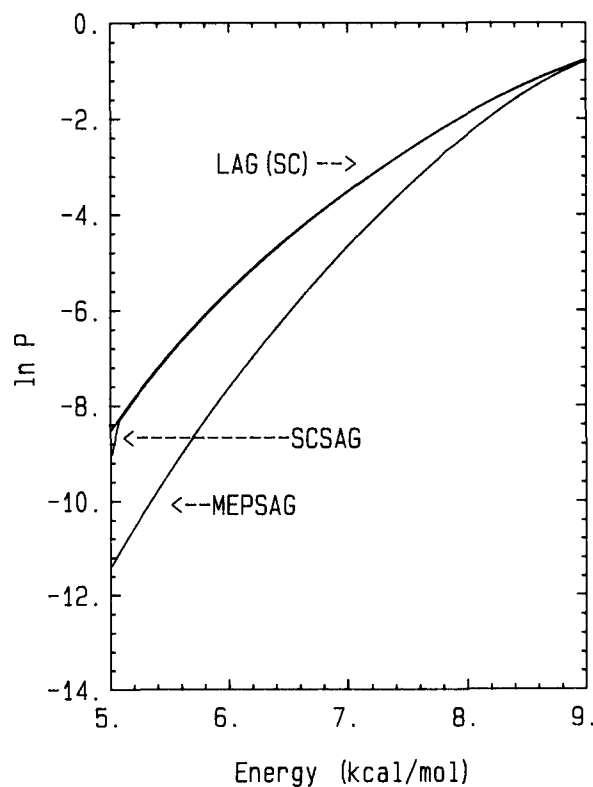


FIG. 4. Same as Fig. 3 for other methods.

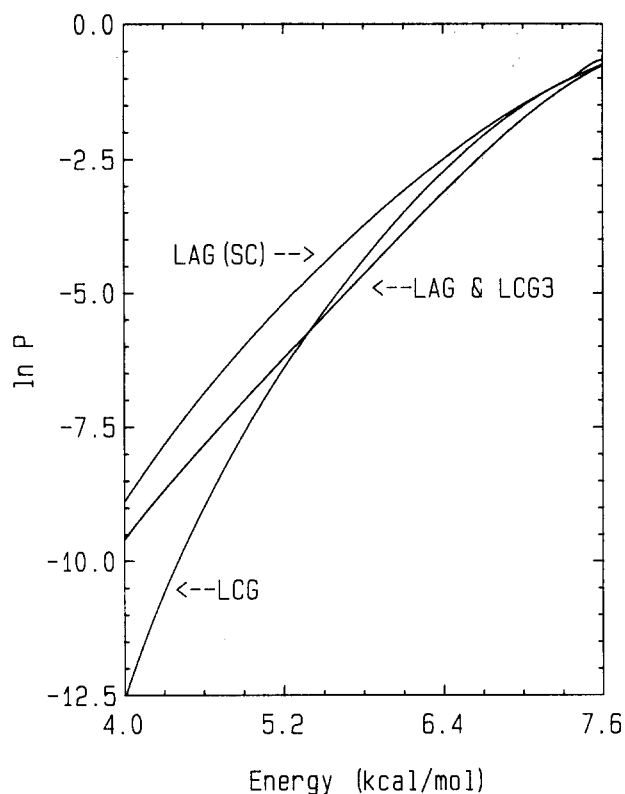


FIG. 5. Same as Fig. 1 for D + BrH → DBr + H.

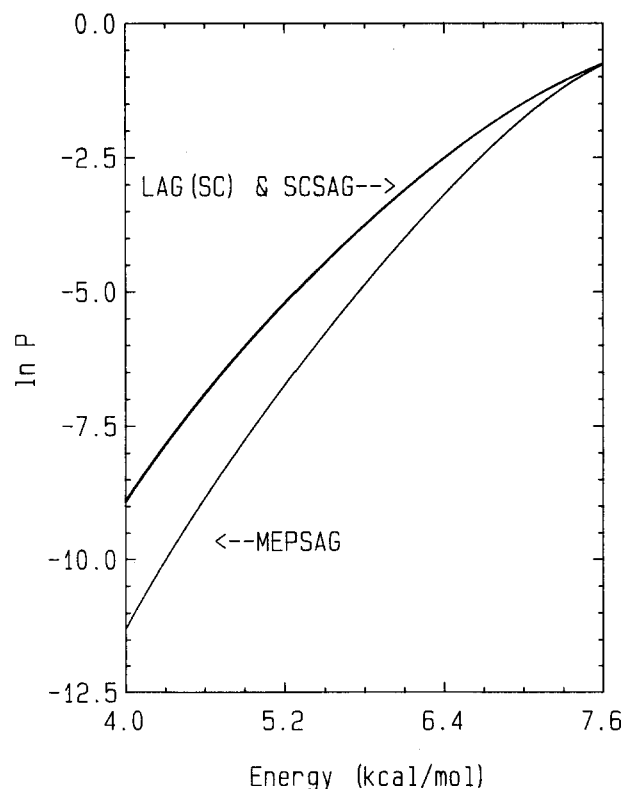


FIG. 6. Same as Fig. 5 for other methods.

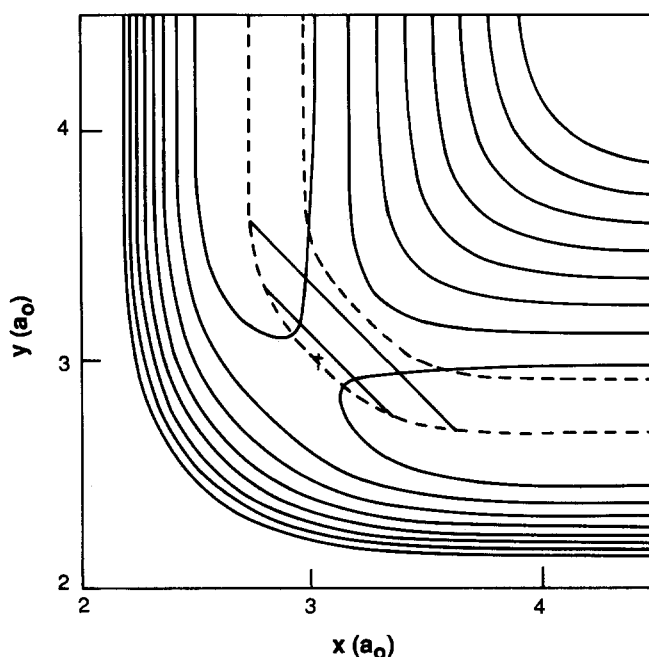


FIG. 7. Contour plot for the collinear $\text{H}' + \text{BrH} \rightarrow \text{H}'\text{Br} + \text{H}$ reaction in mass-scaled Jacobi coordinates with MEP and Marcus-Coltrin paths as dashed curves, saddle point as †, and two tunneling paths. The contour values are 5, 10, 15, 20, 25, 30, 35, and 40 kcal/mol, relative to $\text{H} + \text{HBr}$ ($r = r_e$). The tunneling path closer to the MEP is the LCG3 tunneling path for the collinear reaction at $E = 7.2$ kcal/mol, and the one farther from the MEP is for the three-dimensional reaction at $E = 5.3$ kcal/mol.

IV. DISCUSSION

Tables III–V show similar trends. The conventional TST and the ICVT results are too low compared to the accurate quantal results. The ICVT/MEPSAG results are better than the ICVT and the TST ones but are also significantly too low, i.e., even in a very-small-curvature system, the MEPSAG method underestimates the tunneling contribution significantly, although the error is not as great as when one employs minimum-energy tunneling paths for the case²⁵ of three equal masses.

Tables III–V show that the ICVT/SCSAG and ICVT/LAG(SC) rate constants are very similar to each other in all cases with a difference at a given temperature of at most 10%. Despite this agreement, these results are too high compared to the accurate results for $T \lesssim 500$ K. The ICVT/LAG and the ICVT/LCG3 form another similar pair for all three reactions but the ICVT/LAG and the ICVT/LCG3 methods lead to good agreement with the accurate results. The ICVT/LCG results are too small at low temperature but improve at higher temperature for all three cases.

TABLE VI. Tunneling properties for collinear $\text{H} + \text{BrH}'$.

$T(\text{K})$	E_{mp}^{LAG} (kcal/mol)	$s_{<}$ (a_0)	$s_{>}$ (a_0)	$\alpha_{\text{opt}}^{\text{LAG}}$	$\alpha_{\text{opt}}^{\text{LAG(SC)}}$
300	7.21	−0.24	0.24	1.00	5.4(−1)
400	7.69	−0.15	0.15	1.00	1.00

TABLE VII. Tunneling properties for three-dimensional H + BrH'.

T(K)	E_{mp}^{LAG} (kcal/mol)	$s_{<}$ (a_0)	$s_{>}$ (a_0)	α_{opt}^{LAG}	$\alpha_{opt}^{LAG(SC)}$
200	5.32	-0.79	0.79	1.00	2.0(-2)
300	8.34	-0.23	0.23	1.00	5.7(-1)

The conclusion from the results of Tables III–V is that the LAG method is the best semiclassical tunneling method of the six studied here for reactions (R1) and (R2). Why this is so can be understood by examining the calculations in more detail. The semilog plots in Figs. 1 and 2 for collinear reaction (R1) indicate that the LAG(SC) and the SCSAG transmission probabilities are almost identical, not only when thermally averaged, as discussed above, but also as functions of energy, and similarly for the LAG and the LCG3 tunneling probabilities. Figures 1 and 2 also illustrate that the LAG(SC) and the SCSAG probabilities are the largest at all energies, followed in order by the LAG and LCG3 results, then the LCG, and finally the MEPSAG probabilities. Figures 3–5 show parallel trends and indicate that in all three cases the small-curvature-reference-path methods yield the largest transmission coefficients.

The skew angles for reactions (R1) and (R2) are 89.3 and 89.0 deg, respectively. As mentioned above and as illustrated in Fig. 7 these large skew angles lead to a small-curvature reaction path; however, the small-curvature semiclassical tunneling methods overestimate the rates. The SCSAG method is a generalization of the Marcus–Coltrin path method, and tunneling along the Marcus–Coltrin path provides a visualizable model for the SCSAG tunneling calculation in the present case. The Marcus–Coltrin tunneling path is the path of outer vibrational turning points for the zero-point stretching motion orthogonal to the MEP. This path is shown in the contour plot in Fig. 7 for collinear reaction (R1). This figure indicates that, for the LAG method, the straight-line path connecting the turning points at the dominant tunneling energy never reaches the Marcus–Coltrin path. This unusual circumstance probably occurs only for very small reaction-path curvature. The LCG3 paths in Fig. 7 are close to the MEP path ($\alpha = 0$), so that, independent of the value of α_{opt} , the LAG paths must also be close to the MEP. The optimum α values of the LAG(SC) method at similar energies are all close to $\alpha = 0$ as shown in Tables VI and VII; thus the optimum tunneling path for the LAG(SC) calculations is close to the SC reference path. Figure 7 shows clearly then why the LAG and the LCG3 methods give better results than the small-curvature reaction-path methods, i.e., the optimum tunneling paths occur between the MEP and the SC paths, where they cannot be in Marcus–Coltrin path or LAG(SC) calculations. Thus, the MEP reference allows a more physical choice of tunneling path.

The results obtained indicate that the method which produces the largest transmission coefficient is not necessarily the best method. Although one might have expected such a correlation in the magnitude of the tunneling and the accuracy because of the least-imaginary-action minimization

step in the LAG and the LAG(SC) procedures, the other approximations in the methods are severe enough that the methods do not always underestimate the tunneling.

The good agreement between the ICVT/LAG and ICVT/LCG3 calculations and the accurate quantum results suggests that variational transition state theory with either of these ground-state transmission coefficients is a good approximation for collinear and three-dimensional rate constants, even for a small-curvature system with a large amount of tunneling. These results are very encouraging since in the extension to polyatomics the minimum-energy reference is easier to implement than the small-curvature reference.²⁶ The present results, along with our previous work¹⁰ for cases where the LAG and LAG(SC) methods give similar results, shows that both the LAG and LCG3 methods give reasonable accuracy over a wide range of reaction-path curvatures.

V. SUMMARY OF CONCLUSIONS

A comparative study of several methods for determining transmission coefficients was performed for the H + H'Br → HBr + H' and D + HBr → DBr + H systems on a DIM-3C potential-energy surface. The agreement with available accurate quantum-mechanical results indicates that for both collinear and three-dimensional cases, improved canonical variational transition state theory with a LAG or LCG3 transmission coefficient is accurate within a factor of 1.5 or better for $T > 200$ K. Since the LAG transmission coefficient was found to be the most generally accurate method in previous tests, this is the method we recommend in general, but it is also encouraging for future applications²⁶ that the simpler LCG3 method has such a wide range of applicability. The present new checks on the accuracy of this method give us more confidence about the reliability of LAG tunneling calculations when no accurate quantal dynamics are available. We also gain added confidence in the LCG3 method, which is very easy to implement and has a remarkably wide range of usefulness.

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